INCH-POUND

MIL-C-40004C
26 April 1990
SUPERSEDING
MIL-C-40004B
15 April 1969

MILITARY SPECIFICATION

# CLOTH, KNITTED, COTTON, SIMPLEX

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers two types of simplex knitted cotton cloth.

1.2 Classification. The cloth shall be of the following types and classes as specified (see 6.2):

| | |
|---|---|
| Type I | - 9.5 oz. per sq. yd. |
| Type II | - 7.5 oz. per sq. yd. |
| Class 1 | - White, Shade No. 3028 |
| Class 2 | - Gray, Shade No. 1164 |
| Class 3 | - Gray, Shade No. 1163 |
| Class 4 | - Seal Brown, Shade No. 105 |
| Class 5 | - Gray Beige, Shade No. 270 |
| Class 6 | - Black, Shade No. 3226 |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8305

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-P-1134 - Packaging of Cotton and Cotton-Synthetic Fiber Blend Fabrics (Excluding Duck Fabrics)

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

FEDERAL TRADE COMMISSION

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies are available from the Federal Trade Commission, Pennsylvania Avenue at Sixth Street, N.W., Washington, DC 20580-0001.)



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2.2 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

Chromatic Transference Scale

(Application for copies should be addressed to the American Association of Textile Chemists and Colorists, P.O. Box 12215, Research Triangle Park, NC 27709-2215.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 Standard sample. The finished cloth shall match the standard sample for shade and appearance and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.4).

3.3 Material.

3.3.1 Yarn. The yarn shall be made from cotton which has been carded, combed, and spun into singles yarn.

3.4 Color. The color of the finished cloth shall be as specified (see 6.2). When White (class 1) is specified, the cloth shall be bleached and shall be supplemented with an optical brightner to enhance the appearance of the cloth.

3.4.1 Labile sulfur. The use of dyes and compounds containing elementary sulfur capable of oxidation to sulfuric acid is prohibited. The dyestuff shall be chosen and applied so that the dyed and finished cloth shall contain no more labile sulfur than shown by the standard sample when tested as



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specified in 4.4.3. When no standard sample is available, the dyed and finished cloth shall show no more than a "slight trace" of labile sulfur as defined in the test method specified in 4.4.3.

3.4.2 Matching. The color of the finished cloth shall match the standard sample when viewed under filtered tungsten lamps that approximate artificial daylight and that have a correlated color temperature of 7500 ± 200 K, with illumination of 100 ± 20 foot candles and shall be a good match to the standard sample under incandescent lamplight at 2300 ± 200 K.

3.4.3 Colorfastness. The dyed and finished cloth shall show fastness to laundering and perspiration equal to or better than the standard sample or equal to or better than a rating of "good". The dyed and finished cloth shall show fastness to crocking equal to or better than the standard sample or shall have an AATCC Chromatic Transference Scale rating of not lower than 3.5 or "good", except for class 6 cloth which shall have an AATCC Chromatic Transference Scale rating of not lower than 1.5 or "fair". Testing shall be as specified in 4.4.3.

3.5 Physical requirements. The finished cloth shall conform to the requirements specified in table I when tested as specified in 4.4.3.

TABLE I. Physical requirements

| Type | Weight oz./sq./yd. (min.) | Wales per inch (min.) | Courses per inch (min.) | Bursting strength lbs. (min.) | Stretch, percent Wales (length) (max.) | Courses (width) (min.) | (max.) |
|---|---|---|---|---|---|---|---|
| I | 9.5 | 60 | 42 | 130 | 15 | 40 | 60 |
| II | 7.5 | 66 | 40 | 120 | 20 | 60 | 80 |

3.5.1 Knit. The knit shall be a 48-line Atlas pattern (see 6.5). The cloth shall be knit with nonravelling edges (see 6.5.1).

3.5.2 Width. Unless otherwise specified (see 6.2), the width of the cloth shall be a minimum of 36 inches.

3.6 Finish. The cloth shall be sueded on both sides and have a suppleness equal to or better than the standard sample.

3.7 Non-fibrous materials. The starch and protein content including chloroform-soluble and water-soluble material of the finished cloth shall not exceed 5.0 percent when tested as specified in 4.4.3.

3.8 pH. The pH value of the water extract of the finished cloth shall be no less than 5.0 nor more than 8.5 when tested as specified in 4.4.3.



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3.9 Dimensional stability. There shall be no more than 12.0 percent shrinkage in the direction of the length (wale direction) or 3.0 percent in the width (course direction) when tested as specified in 4.4.3. The preshrinkage process used shall not be identified by name or tradmark either on the cloth, ticket or package.

3.10 Length and put-up. For Government procurement only, unless otherwise specified (see 6.2), the cloth shall be furnished in continuous lengths, each not less than 35 yards. Each length shall be put-up on full width rolls as specified in 5.1.

3.11 Fiber identification. Each roll shall be labeled, ticketed or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act.

3.12 Workmanship. The finished cloth shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.



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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.2.1 through 4.4.2.3 and tested as specified in 4.4.3.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 End item examination.

4.4.2.1 Yard-by-yard examination. The cloth shall be examined for the defects listed in table II. All defects found shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the cloth, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one for each wale-wise (length-wise) yard or fraction thereof in which it occurs. The lot size shall be expressed in units of yards. The sample unit shall be 1 linear yard. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 6.5 for major defects and 10 for total defects (major and minor combined). The number of rolls from which the sample yardage is to be selected shall be in accordance with table III. The sample yardage shall be apportioned equally among the selected rolls.

TABLE II. End item visual defects

| Defects | Classification Major | Minor |
|---|---|---|
| Hole, cut, tear, any | 101 | |
| Run or end out | 102 | |
| Drop stitch, tuck stitch, pulled stitch or needle streak | | 201 |
| Mend | 103 | |
| Abrasion mark greater than 1 inch | 104 | |
| Thin or uneven cloth 1/ | 105 | |
| Course or fine yarn 1/ | 106 | |
| Tight twist yarn | | 202 |
| Slubs or gouts greater than 1 inch | | 203 |
| Specks greater than 1/8 inch | | 204 |



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TABLE II. End item visual defects (cont'd)

| Defects | Classification Major | Minor |
|---|---|---|
| Knots extending above the surface of the cloth 1/ | | 205 |
| Kinks or untrimmed ends | | 206 |
| Crease or wrinkle, (embedded) | | 207 |
| Stretch (stitch tension) knitting tension too tight or too loose | 107 | |
| Shade bar 1/ | | 208 |
| Spot or stain greater than 1/8 inch | 108 | |

1/ Clearly visible at normal inspection distance (3 feet).

4.4.2.2 Overall examination. The cloth shall be examined for the defects listed below. Each defect listed shall be counted not more than once in each roll examined. The sample size (number of rolls) and the acceptance number (maximum number of defects acceptable) shall be in accordance with table III. Each roll shall be examined over its entire length.

Defects

Preshrinkage process identified by name or trademark on the cloth or ticket.
Degree and character of finish not equal to standard sample.
Degree and character of sueding not equal to standard sample.
Objectionable odor.
Overall uncleanness.
Uneven knitting throughout the roll.
Off shade, shaded end to end, side to side, side to center or throughout the roll.
Mottled, streaky or cloudy in appearance.
Edges frayed, ravelled, curled, rolled, folded, scalloped or uneven continuously or intermittently throughout the roll.
Not labeled in accordance with Textile Fiber Products Identification Act.
Overall width less than specified.



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TABLE III. Sample size

| Lot size (yards) | Sample size (rolls) | Acceptance number |
|---|---|---|
| 1200 or less 1/ | 3 | 0 |
| 1201 up to and including 3,200 | 5 | 0 |
| 3,201 up to and including 10,000 | 8 | 0 |
| 10,001 up to and including 35,000 | 13 | 0 |
| 35,001 up to and including 150,000 | 20 | 1 |
| 150,001 and over | 32 | 2 |

1/ If a lot contains fewer than three rolls, each roll in the lot shall be examined.

4.4.2.3 Length examination. During the yard-by-yard examination, each roll in the sample shall be examined for length. Any length found to be less than the minimum specified or more than 2 yards less than the length marked on the roll ticket shall be considered a length defect. If the total of the actual lengths of the rolls in the sample is less than the total of the lengths marked on the roll tickets, the lot shall be rejected.

4.4.3 End item testing. The cloth shall be tested for the characteristics listed in table IV. The methods of testing specified in FED-STD-191 wherever applicable and as listed in table IV shall be followed. All test reports shall contain the individual values utilized in expressing the final results. The sample unit shall be 3-1/2 continuous yards full width of the finished cloth. The lot shall be rejected if one or more sample units fail to meet any requirement specified. The sample size shall be in accordance with the following:

| Lot size (yards) | Sample size (sample units) |
|---|---|
| 800 or less | 2 |
| 801 up to and including 22,000 | 3 |
| 22,001 and over | 5 |

TABLE IV. End item tests

| Characteristic | Requirement paragraph | Test method |
|---|---|---|
| Cotton identification | 3.3.1 | 1/ |
| Optical brightner (class 1) | 3.4 | 1/ |
| Labile sulfur content | 3.4.1 | 2020 |



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TABLE IV. End item tests (cont'd)

| Characteristic | Requirement paragraph | Test method |
|---|---|---|
| Colorfastness to: | | |
| Laundering | 3.4.3 | 5614 2/ |
| Perspiration | 3.4.3 | 5680 |
| Crocking | 3.4.3 | 5651 |
| Weight, oz./sq. yd. | 3.5 | 5041 |
| Texture | | |
| Wales per inch | 3.5 | 5070 |
| Courses per inch | 3.5 | 5070 |
| Bursting strength, lbs. | 3.5 | 5120 |
| Stretch, percent | | |
| Wale direction | 3.5 | 4.5.1 |
| Course direction | 3.5 | 4.5.1 |
| Knit | 3.5.1 | 1/ |
| Nonfibrous material | 3.7 | 2611 |
| pH | 3.8 | 2811 |
| Dimensional stability | 3.9 | 5550 |

1/ A certificate of compliance shall be submitted and will be acceptable for the stated requirement.

2/ For classes 2 and 3 cloths, the temperature of the water during the test shall be 160° ± 2°F.

4.4.4 Packaging inspection. The sampling and inspection of the preservation, packaging, and container marking shall be in accordance with the requirements of PPP-P-1134.

4.5 Methods of inspection.

4.5.1 Stretch test.

4.5.1.1 Preparation of specimens. One specimen each from the length (wales) and width (courses) direction shall be tested from each sample unit. Each specimen shall be a rectangle 4 inches by not less than 5 inches. The



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long dimension shall be parallel to the length (wales) of the cloth for length tests and parallel to the width (courses) for width tests. The test specimens shall be taken no nearer the selvage than 1/10 the width of the cloth. Two parallel lines, 3 inches apart shall be drawn at right angles to the long dimension of the specimen. In addition, a line shall be drawn parallel to the long dimension of the specimen and 1 inch in from either edge for properly aligning the specimen in the jaws of the testing apparatus.

4.5.1.2 Test apparatus. The test device (see figure 1) shall be an apparatus wherein the specimen is held between two clamps. The design of the upper clamp shall be such that one gripping surface or jaw shall be an integral part of the rigid frame of the clamp. The face of the movable jaw shall measure 1 inch by 1 inch and the face of the fixed jaw shall measure 1 inch by 1-1/2 or more inches with the long dimension perpendicular to the direction of application of the load. The lower clamp shall have similar jaws and, when weight is attached, shall weigh a total of 6 pounds. The jaws shall have smooth gripping surfaces, sufficiently flat and parallel to prevent slipping of the specimen during the test.

4.5.1.3 Procedure. The specimen shall be placed symmetrically in the clamps of the apparatus with the long dimension parallel and the short dimension at right angles to the direction of the application of the load. Using the specified guidelines, the specimen shall be aligned with one outside edge of the front jaws of each clamp. The proper weight shall be added to insure a total of 6 pounds (including the weight of the lower jaws). The load shall be applied for 15 seconds and the distance between jaws measured by a pair of dividers or other suitable instrument. The elongation at the required load shall be expressed as the percent increase in the length of the specimen between the jaws. The percent stretch shall be calculated as follows:

$$\text{Percent stretch} = \frac{\text{Stretch dimension} - \text{Original 3-inch dimension}}{\text{Original dimension (3 inches)}} \times 100$$

4.5.1.4 Report. The results obtained shall be reported separately to the nearest 1.0 percent.

5. PACKAGING

5.1 Put-up and preservation. Put-up and preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Levels A and Commercial. The cloth shall be put-up and preserved in accordance with the applicable requirements of PPP-P-1134.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.2.1 Levels A, B, and Commercial. The cloth shall be packed in accordance with the applicable requirements of PPP-P-1134.



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5.3 Marking. In addition to any special marking required by the contract or purchase order, shipments shall be marked in accordance with the applicable requirements of PPP-P-1134.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cloth is intended for use in the fabrication of gloves, scarfs and lining of protective headgear, and earphones and other personal equipment.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Type and class (color) required (see 1.2 and 3.4).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When a first article is required (see 3.1, 4.3, and 6.3).
e. Width required, if other than specified (see 3.5.2).
f. Length required, if other than specified (see 3.10).
g. Levels of preservation and packing (see 5.1 and 5.2).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Sample. For access to samples, address the contracting activity issuing the invitation for bids or request for proposal.

6.5 Knit. The 48-line Atlas pattern is produced when the warp ends continue to move in the same direction for 24 consecutive courses after which the direction of the guide bar movement is reversed and the warp ends traverse 24 courses in the opposite direction. A faint horizontal line occurs at each half repeat where the guide bars change direction. This represents one full repeat of the pattern.

6.5.1 Knitting machine. The type I and type II cloth has been successfully knit on a 32 gauge, two-needle bar warp knitting machine (simplex).



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6.6 Subject term (key word) listing.

Gloves
Lining material
Personal equipment

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:

Army - GL
Navy - NU
Air Force - 11

Review activities:

Army - MD
Navy - MC
Air Force - 82, 99
DLA - CT

Preparing activity:

Army - GL

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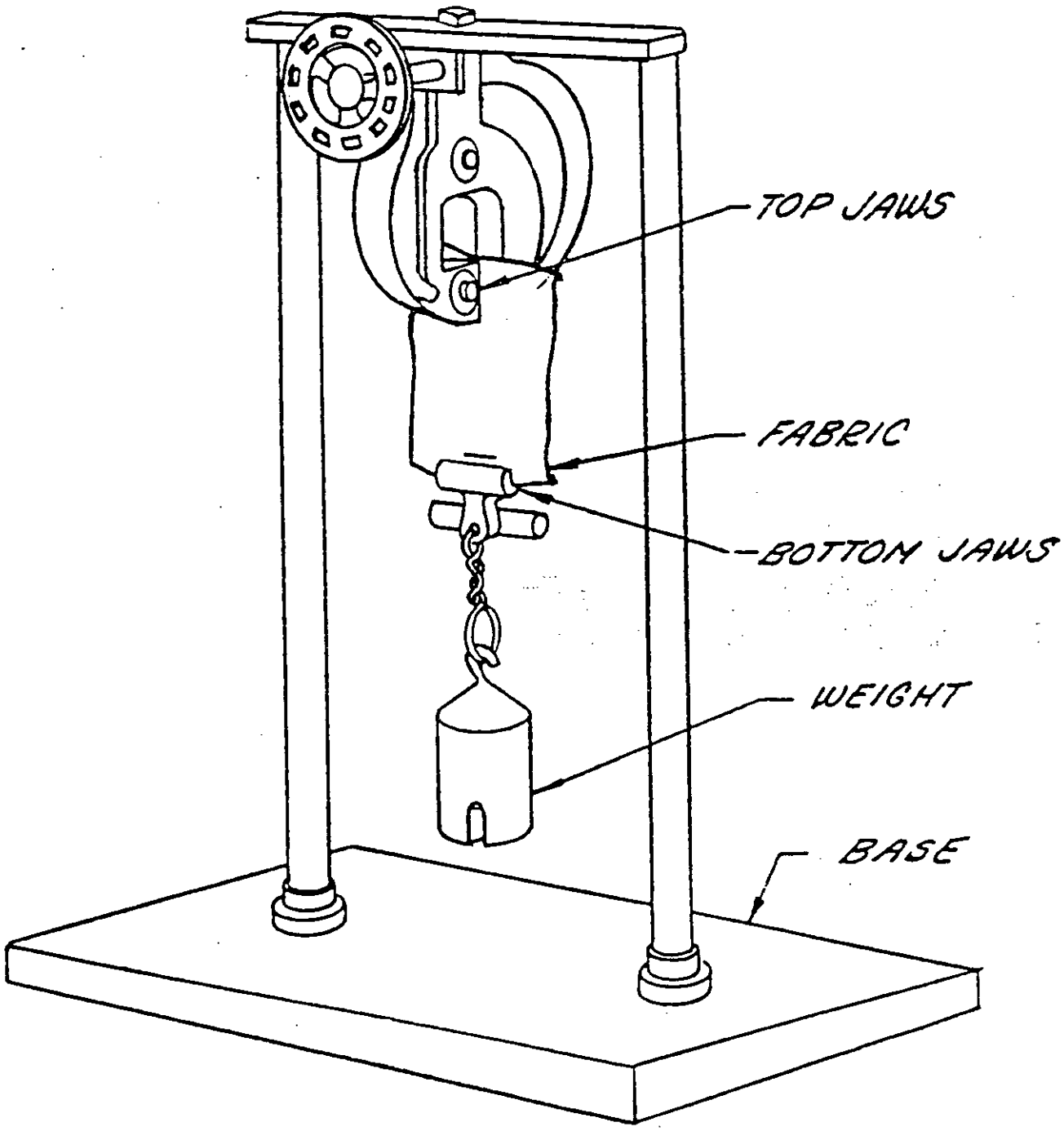



FIGURE 1. *Fabric stretch apparatus.*

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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

## INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER | 2. DOCUMENT DATE (YYMMDD) |
|---|---|---|
| | MIL-C-40004C | 90/04/26 |

3. DOCUMENT TITLE
CLOTH, KNITTED, COTTON, SIMPLEX

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON (If applicable) | 7. DATE SUBMITTED (YYMMDD) |

8. PREPARING ACTIVITY

| a. NAME | b. TELEPHONE (Include Area Code) | |
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| U.S. Army Natick RD&E Center | (1) Commercial<br>508-651-5221 | (2) AUTOVON<br>256-5221 |
| c. ADDRESS (Include Zip Code)<br>Commander, U.S. Army Natick RD&E Center<br>ATTN: STRNC-ES<br>Natick, MA 01760-5014 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 | |

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